



Mathematics Part 1 — Cbse Periodic Test (Pt) — 40 Marks

ANSWER KEY — FOR GRADING

Subject: Mathematics Part 1 | Class: Class 9

Total Marks: 40 | Time Allowed: 1h 30m | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 8 questions. Each carries 1 mark(s).		
1	What is the distance between the points (3, 5) and (7, 5)? (A) 2 (B) 12 (C) 4 (D) $\sqrt{32}$ Answer: Correct Option: (C) ► Why? C. The distance between points (x_1, y) and (x_2, y) is the absolute value $x_2 - x_1$. Here, $7 - 3 = 4$. [Ch 1: Orienting Yourself: The Use of Coordinates]	1
2	The ancient Egyptian formula for the area of a circle was $A \approx (8d/9)^2$, where d is the diameter. This can also be expressed as: (A) $(64/81)r^2$ (B) $(128/81)r^2$ (C) $(256/81)r^2$ (D) $(32/81)r^2$ Answer: Correct Option: (C) ► Why? C. Since $d = 2r$, we get $A \approx ((8 \cdot 2r)/9)^2 = (16r/9)^2 = 256/81r^2$. [Ch 6: Measuring Space: Perimeter and Area]	1



Q.NO.	QUESTIONS	MARKS
3	The belief that if a random event happens multiple times in a row, the opposite outcome is more likely to happen next, is known as: (A) Law of Large Numbers (B) Gambler's Fallacy (C) Subjective Probability (D) Sampling Error Answer: Correct Option: (B) ► Why? B. Gambler's Fallacy is the common misunderstanding that past random events affect the likelihood of future independent events, such as believing a coin has a higher chance of tails after several heads. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	1
4	When we change the values of b while keeping the value of a fixed in the equation $y = ax + b$, what happens to the lines? (A) They intersect at the origin (B) They shift but remain parallel to the original line (C) They become perpendicular (D) They have different slopes Answer: Correct Option: (B) ► Why? B. If we change the values of b keeping the value of a fixed, the lines shift but remain parallel to the original line since they have equal slopes but different y-intercepts. [Ch 2: Introduction to Linear Polynomials]	1
5	Assertion (A): Linear decay describes a pattern in which a quantity decreases by a fixed amount over equal intervals. Reason (R): In a linear relationship represented by $y = ax + b$, the constant b represents the slope, and a straight line representing linear decay has a positive slope. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (C) ► Why? C. The assertion is true because linear decay describes a pattern where a quantity decreases by a fixed amount over equal intervals. The reason is false because in $y = ax + b$, the constant b is the y-intercept, not the slope, and linear decay is represented by a straight line with a negative slope, not a positive slope. [Ch 2: Introduction to Linear Polynomials]	1



Q.NO.	QUESTIONS	MARKS
6	If one pushes a chord away from the centre, at some stage the chord becomes a single point. What is the distance of the centre from this point? (A) Zero (B) Equal to the chord length (C) Half the radius (D) The radius Answer: Correct Option: (D) ► Why? D. If the chord becomes a single point, its length is zero, and the distance of the centre from this chord is the radius. [Ch 5: I'm Up and Down, and Round and Round]	1
7	Since $x^4 - y^4 = (x^2 - y^2)(x^2 + y^2)$, which of the following must be a factor of $x^4 - y^4$? (A) $x + y$ only (B) $x^2 + y^2$ only (C) $x - y$ (D) $x^2 - y^2$ only Answer: Correct Option: (C) ► Why? C. Since $x^2 - y^2 = (x - y)(x + y)$, the expression $x^4 - y^4$ can be written as $(x - y)(x + y)(x^2 + y^2)$, making $x - y$ a factor. [Ch 4: Exploring Algebraic Identities]	1
8	Assertion (A): The point $(3, -5)$ lies in Quadrant II. Reason (R): Points in Quadrant II have a negative x-coordinate and a positive y-coordinate. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (D) ► Why? D. The assertion is false because the source explicitly states that point S $(3, -5)$ is in Quadrant IV, not Quadrant II, since it has a positive x-coordinate and a negative y-coordinate. The reason is true as it correctly states the sign convention for Quadrant II. [Ch 1: Orienting Yourself: The Use of Coordinates]	1
Section B Attempt all 4 questions. Each carries 2 mark(s).		
9	The cost of a journey is given by $C(d) = 100 + 60d$. By what constant amount does the cost increase per km, and what does this represent?	2



Q.NO.	QUESTIONS	MARKS
	Answer: The cost increases by a fixed amount of ₹60 for every km. Since the value of the cost function increases by a constant amount over equal intervals, this represents linear growth. ► Why? A constant increase in the dependent variable (cost) for equal increments of the independent variable (distance) defines linear growth. [Ch 2: Introduction to Linear Polynomials]	
10	In a survey of 50 students, 15 students said they liked football. The number of students who like football is 15, and the ____ (frequency/relative frequency) is ____ (fill in the fraction or decimal). Answer: relative frequency, 0.3 (or 15/50) ► Why? According to the provided context, the ratio of the number of times an event occurs to the total number of trials is called the relative frequency (e.g., relative frequency of rolling a 4 is $8/50 = 0.16$). Here, the number of students who like football is 15 out of a total of 50 students. Thus, the relative frequency is calculated as $15/50$, which simplifies to 0.3. ► Step-by-step: - Step 1: Identify the term for the ratio of observed occurrences to total trials, which is relative frequency. - Step 2: Calculate the relative frequency using the formula: Relative frequency = (Number of students who like football) / (Total number of students) = $15 / 50 = 0.3$. [Ch 7: The Mathematics of Maybe: Introduction to Probability]	2
11	In which quadrant does a point with a negative x-coordinate and a positive y-coordinate lie? Answer: A point with a negative x-coordinate and a positive y-coordinate lies in Quadrant II. This is one of the four quadrants formed by the intersection of the coordinate axes. ► Why? This requires understanding and applying the sign conventions for the quadrants as explicitly laid out in the source material. [Ch 1: Orienting Yourself: The Use of Coordinates]	2
12	State Brahmagupta's rule regarding the product of two debts and express it in modern notation. Answer: Brahmagupta's Laws stated that 'the product of two debts is a fortune,' meaning that multiplying two negative numbers yields a positive result. In modern notation, this is expressed as $(-) \times (-) = (+)$. ► Why? The source explicitly mentions Brahmagupta's rule that the product of two debts is a fortune, equivalent to $(-)\times(-) = (+)$. [Ch 3: The World of Numbers]	2
Section C Attempt 3 out of 4.		



Q.NO.	QUESTIONS	MARKS
13	The village playground is shaped as a square of side 40 metres. A path of width s metres is created around the playground for people to walk. Find an expression for the area of the path in terms of s. Answer: $160s + 4s^2$ square metres ► Why? The area of the path is found by subtracting the area of the inner square playground from the area of the outer square (playground + path). The inner square has an area of 40^2. The outer square has a side length of $40 + 2s$, making its area $(40 + 2s)^2$. Using the difference of squares identity, the path's area simplifies to $160s + 4s^2$. ► Step-by-step: - Step 1: Area of the inner square playground = $40^2 = 1600$ sq. metres. - Step 2: Side of the outer square including the path = $40 + s + s = 40 + 2s$ metres. Area of outer square = $(40 + 2s)^2$ sq. metres. - Step 3: Area of the path = Area of outer square - Area of inner square = $(40 + 2s)^2 - (40)^2$. - Step 4: Using the identity $a^2 - b^2 = (a+b)(a-b)$, we get $(40 + 2s + 40)(40 + 2s - 40) = (80 + 2s)(2s) = 160s + 4s^2$ sq. metres. [Ch 4: Exploring Algebraic Identities]	3
	OR Use the Baudhāyana-Pythagoras theorem to show why Theorem 6 must be true. Answer: Given: Circle with centre C, chords AB = FG. E and H are midpoints of AB and FG respectively. CE and CH are perpendiculars from C to AB and FG. To show: CE = CH. By the Baudhāyana-Pythagoras theorem in right triangles CEA and CHF, we have $CA^2 = CE^2 + AE^2$ and $CF^2 = CH^2 + FH^2$. Since CA and CF are radii of the same circle, $CA = CF$, which means $CA^2 = CF^2$. Therefore, $CE^2 + AE^2 = CH^2 + FH^2$. Since E and H are midpoints of equal chords AB and FG, $AE = FH$, which means $AE^2 = FH^2$. Substituting this, we get $CE^2 = CH^2$, which implies $CE = CH$. Thus, chords of equal length are equidistant from the centre. ► Why? The question asks to prove Theorem 6 using the Baudhāyana-Pythagoras theorem instead of congruence. Theorem 6 states that equal chords are equidistant from the centre. By dropping perpendiculars from the centre to the chords, we form two right triangles. Applying the Pythagoras theorem to both, and equating the squares of the radii (since they are equal), we can substitute the equal half-chord lengths to prove that the squares of the perpendicular distances are equal, hence the distances themselves are equal. ► Step-by-step: - Step 1: Identify right triangles formed by dropping perpendiculars from centre C to chords AB and FG, forming $CE \perp AB$ and $CH \perp FG$. - Step 2: Apply Baudhāyana-Pythagoras theorem: $CA^2 = CE^2 + AE^2$ and $CF^2 = CH^2 + FH^2$. - Step 3: Equate CA^2 and CF^2 since CA and CF are radii: $CE^2 + AE^2 = CH^2 + FH^2$. - Step 4: Substitute $AE^2 = FH^2$ (since E and H are midpoints of equal chords AB and FG). - Step 5: Conclude $CE^2 = CH^2$, which gives $CE = CH$. [Ch 5: I'm Up and Down, and Round and Round]	
14	Find the 10 th term of the GP: 5, 25, 125,	3



Q.NO.	QUESTIONS	MARKS
	Answer: Given: GP is 5, 25, 125, ... First term a = 5, Common ratio r = 25/5 = 5 Formula: $t_n = ar^{n-1}$ Substitution: $t_{10} = 5 \times (5)^{10-1} = 5 \times 5^9 = 5^{10}$ Calculation: $5^{10} = 9765625$ Answer: $\therefore$ The 10th term of the GP is 9765625. A geometric progression (GP) is a sequence where each term after the first is obtained by multiplying the previous term by a fixed constant called the common ratio r. The n^{th} term is given by the formula $t_n = ar^{n-1}$. By identifying the first term a and the common ratio r, we can calculate any term in the sequence directly, demonstrating the power of the explicit formula for efficient computation. ► Why? By identifying the first term and common ratio from the given GP, the explicit formula for the n^{th} term is used to directly compute the 10th term. ► Step-by-step: 1. Step 1: Identify the first term $a = 5$ and common ratio $r = 5$ from the given GP. 2. Step 2: Apply the GP explicit formula $t_n = ar^{n-1}$ for the 10th term. 3. Step 3: Substitute the values to calculate $t_{10} = 5 \times 5^9 = 5^{10}$. [Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]	
15	What value of π was provided by Āryabhaṭa in 499 CE, and what profound insight did he express about this value using the word 'asanna'? Also, write Mādhava's exact formula for π. Answer: In 499 CE, Āryabhaṭa provided the value $62832/20000 = 3.1416$ for π. Crucially, he described this value as asanna, meaning 'approaching' or 'approximate'. This was a profound insight because it suggested that the ratio of a circle's circumference to its diameter could not be given exactly as one simple fraction. Mādhava's exact formula for π is given as an infinite series: $\pi = 4(1 - 1/3 + 1/5 - 1/7 + \dots)$ This formula provides a way to calculate π exactly, confirming Āryabhaṭa's insight that it is not a simple fraction.	3



Q.NO.	QUESTIONS	MARKS
	► Why? Aryabhata recognized the irrational nature of π by calling his fraction 'asanna' (approximate). Madhava later provided an exact infinite series representation, proving π cannot be a simple fraction. ► Step-by-step: - Step 1: State Āryabhaṭa's value for π as $62832/20000$ or 3.1416. - Step 2: Define 'asanna' as 'approaching' or 'approximate', explaining that it implies π cannot be exactly represented as a simple fraction. - Step 3: Write down Mādhava's infinite series formula for π. [Ch 6: Measuring Space: Perimeter and Area]	
Section D Attempt 2 out of 3.		
16	A gym charges a fixed monthly fee and an additional cost per hour for using the badminton court. When the court is used for 10 hours, the bill is ₹800, and when it is used for 15 hours, the bill is ₹1100. The monthly bill y depends on the hours of use x by the relation $y = ax + b$. Find the values of a and b. Identify the slope and the y-intercept of the line, and explain what they represent in this context.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Given: The monthly bill y depends on hours of use x by the relation $y = ax + b$. When $x = 10$, $y = 800$ and when $x = 15$, $y = 1100$. Substitution: $800 = 10a + b$ $1100 = 15a + b$ Calculation: Subtracting the first equation from the second: $1100 - 800 = 15a - 10a$ $300 = 5a$ $a = 60$ Substituting $a = 60$ into $10a + b = 800$: $10(60) + b = 800$ $600 + b = 800$ $b = 200$ Equation: $y = 60x + 200$ Slope and y-intercept: In the equation $y = ax + b$, a represents the slope and b represents the y-intercept. Here, slope $a = 60$ and y-intercept $b = 200$. Contextual Meaning: The slope $a = 60$ represents the additional cost per hour for using the badminton court (₹60 per hour). The y-intercept $b = 200$ represents the fixed monthly fee charged by the gym (₹200), which is the bill when the court is used for 0 hours. ∴ The values are $a = 60$ and $b = 200$. The slope is 60 and the y-intercept is 200.	



Q.NO.	QUESTIONS	MARKS
	► Why? The values of a and b are found by solving the system of linear equations formed from the given conditions; a represents the variable cost (slope) and b represents the fixed cost (y-intercept). ► Step-by-step: - Step 1: Substitute the given hours and bill values into the equation $y = ax + b$ to form two equations. - Step 2: Solve the system of equations to find the values of a and b. - Step 3: Identify that a is the slope and b is the y-intercept. - Step 4: Interpret the slope as the cost per hour and the y-intercept as the fixed monthly fee in the context of the gym. [Ch 2: Introduction to Linear Polynomials]	
	OR Brahmagupta introduced the concepts of "fortunes" (Dhana) and "debts" (Riña) to represent positive and negative numbers. Using these real-world concepts, explain why: (i) the product of two debts is a fortune, and (ii) subtracting a negative number is the same as adding a positive number. Provide a numerical example for each. Answer: Brahmagupta grounded his mathematics in the reality of commerce by introducing fortunes (Dhana) to represent positive numbers (wealth or assets) and debts (Riña) to represent negative numbers (debts). This real-world connection helps explain the rules of integer arithmetic logically. (i) The product of two debts is a fortune: A negative number represents a debt, and multiplying by a negative number represents the removal of that debt. If someone takes away four of your debts, each worth ₹3 (which is -3), you are effectively ₹12 richer. Mathematically, taking away a debt is multiplying by a negative: $(-3) \times (-4) = +12$ The removal of a debt increases your wealth, hence a fortune. (ii) Subtracting a negative number is the same as adding a positive number: If you have a wealth of ₹10 and someone takes away a debt of ₹5 (represented as -5), your net wealth increases. The removal of a -5 debt means you gain ₹5. This is expressed as: $10 - (-5) = 15$ Subtracting a debt adds to your fortune. Thus, Brahmagupta's rules perfectly mirror real-life financial transactions, showing that arithmetic operations on integers are logical and consistent.	



Q.NO.	QUESTIONS	MARKS
	► Why? The question requires interpreting the abstract rules of integer multiplication and subtraction through Brahmagupta's real-world analogies of wealth (fortunes) and loans (debts). ► Step-by-step: - Step 1: Define Brahmagupta's concepts of fortunes (positive) and debts (negative). - Step 2: Explain that multiplying two negatives means removing a debt, which increases wealth (product is positive). Give the numerical example $(-3) \times (-4) = 12$. - Step 3: Explain that subtracting a negative means taking away a debt, which adds to wealth (same as adding a positive). Give the numerical example $10 - (-5) = 15$. - Step 4: Conclude by summarizing how these rules reflect real-world commerce. [Ch 3: The World of Numbers]	
17	A city has two main roads which cross each other at the centre of the city. These two roads are along the North-South (N-S) direction and East-West (E-W) direction. All the other streets of the city run parallel to these roads and are 200 m apart. There are 10 streets in each direction. (i) Using 1 cm = 200 m, draw a model of the city in your notebook. Represent the roads/streets by single lines. (ii) There are street intersections in the model. Each street intersection is formed by two streets—one running in the N-S direction and another in the E-W direction. Each street intersection is referred to in the following manner: If the second street running in the N-S direction and 5th street in the E-W direction meet at some crossing, then we call this street intersection (2, 5). Using this convention, find: (a) how many street intersections can be referred to as (4, 3). (b) how many street intersections can be referred to as (3, 4). Answer: (i) Draw two perpendicular lines intersecting at the center (Origin). Using 1 cm = 200 m, draw 5 parallel lines at 1 cm intervals on both sides of the N-S road and 5 parallel lines at 1 cm intervals on both sides of the E-W road.	5



Q.NO.	QUESTIONS	MARKS
	► Why? The city model represents a 2-D Cartesian coordinate system where the main roads act as the x-axis and y-axis, crossing at the origin (0,0). Because distances to the right and upwards are positive, and distances to the left and downwards are negative, a specific street number (like the 4th street) exists on both the positive and negative sides of the origin. Therefore, any intersection (a, b) will have four possible locations in the grid: (a, b), (a, -b), (-a, b), (-a, -b) ► Step-by-step: 1. Step 1: Recognize the city model as a 2-D Cartesian coordinate system with the center crossing as the origin (0,0). 2. Step 2: Understand that N-S streets represent x-coordinates (positive to the East, negative to the West) and E-W streets represent y-coordinates (positive to the North, negative to the South). 3. Step 3: For intersection (4, 3), the 4th N-S street can be $x = 4$ or $x = -4$. The 3rd E-W street can be $y = 3$ or $y = -3$. The four possible intersections are: (4, 3), (4, -3), (-4, 3), (-4, -3) 4. Step 4: For intersection (3, 4), the 3rd N-S street can be $x = 3$ or $x = -3$. The 4th E-W street can be $y = 4$ or $y = -4$. The four possible intersections are: (3, 4), (3, -4), (-3, 4), (-3, -4) [Ch 1: Orienting Yourself: The Use of Coordinates]	
Section E Attempt 1 out of 1.		
18	Case Study: Room Layout with Coordinates Reiaan is planning his room layout on a coordinate plane. The room door extends from $D_1 (9.5, 0)$ to $R_1 (11.5, 0)$. The bathroom door extends from $B_1 (0, 1.5)$ to $B_2 (0, 4)$. A rectangular study table has three of its feet at the points (8, 9), (11, 9) and (11, 7). Reiaan wants to verify the dimensions of each item and check whether the bathroom door is wider or narrower than the room door. He also needs to find the position of the fourth foot of the table to complete the layout. (i) What is the width of the room door? (ii) What are the coordinates of the fourth foot of the rectangular study table? (iii) Find the width of the bathroom door. Is the bathroom door narrower or wider than the room door? (iv) Both B_1 and B_2 have x-coordinate equal to 0. What does this indicate about the position of the bathroom door with respect to the y-axis? Answer: (i) Width = $11.5 - 9.5 = 2$ units (ii) (8, 7). Since the table is rectangular, the fourth foot must have the same x-coordinate as (8, 9) and the same y-coordinate as (11, 7). (iii) Width of bathroom door = $4 - 1.5 = 2.5$ units. The bathroom door is wider than the room door ($2.5 > 2$). (iv) The bathroom door lies along the y-axis, since both its endpoints have x-coordinate 0, meaning their perpendicular distance from the y-axis is zero.	5



Q.NO.	QUESTIONS	MARKS
	► Why? i) $\text{Width} = 11.5 - 9.5 = 2 \text{ units}$ ii) (8, 7). Since the table is rectangular, the fourth foot must have the same x-coordinate as (8, 9) and the same y-coordinate as (11, 7). iii) $\text{Width of bathroom door} = 4 - 1.5 = 2.5 \text{ units}$. The bathroom door is wider than the room door ($2.5 > 2$). iv) The bathroom door lies along the y-axis, since both its endpoints have x-coordinate 0, meaning their perpendicular distance from the y-axis is zero. [Ch 1: Orienting Yourself: The Use of Coordinates]	

